SET-4

Series %BAB%/C

प्रश्न-पत्र कोड
Q.P. Code **65/B/6**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **7** हैं ।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **14** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
- Please check that this question paper contains **7** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **14** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

गणित

(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

MATHEMATICS

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 2 घण्टे | *अधिकतम अंक : 40*
Time allowed : 2 hours | *Maximum Marks : 40*

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains* ***three*** *sections –* ***Section A, B*** *and* ***C****.*

(ii) *Each section is* ***compulsory****.*

(iii) ***Section A*** *has* ***6*** *short answer type I questions of* ***2*** *marks each.*

(iv) ***Section B*** *has* ***4*** *short answer type II questions of* ***3*** *marks each.*

(v) ***Section C*** *has* ***4*** *long answer type questions of* ***4*** *marks each.*

(vi) *There is an internal choice in some questions.*

(vii) *Question no.* ***14*** *is a case-study based question with 2 sub-parts of* ***2*** *marks each.*

SECTION A

Questions number ***1*** *to* ***6*** *carry* ***2*** *marks each.*

1. Find the general solution of the differential equation : *2*

$$\frac{dy}{dx} = (1 + x^2)(1 + y^2)$$

2. (a) Find : $\int \frac{x^3}{\sqrt{1 - x^8}}\, dx$ *2*

OR

(b) Find : $\int \frac{e^x \cdot x}{(x + 1)^2}\, dx$ *2*

3. Two cards are drawn at random without replacement from a well-shuffled deck of 52 playing cards. Find the probability of getting both cards of the same colour. *2*

4. A bag contains 2 white, 2 red and 3 blue balls. Two balls are drawn at random from the bag one-by-one with replacement. Find the probability distribution of the number of white balls. *2*

5. Find the value of λ for which the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular to each other, where $\vec{a} = 2\hat{i} - \hat{j} + \lambda\hat{k}$ and $\vec{b} = 3\hat{i} + \hat{j} - 2\hat{k}$. *2*

6. Find the direction cosines of a line whose vector equation is given as $\vec{r} = (1 - \lambda)\hat{i} + (2\lambda - 1)\hat{j} + (2\lambda + 3)\hat{k}$. *2*

SECTION B

*Questions number **7** to **10** carry **3** marks each.*

7. Evaluate : $\displaystyle\int_{-4}^{4} |x + 2|\, dx$ *3*

8. (a) Find the particular solution of the differential equation : *3*

$$(x \cos^2 (y/x) - y)\, dx + x\, dy = 0, \text{ given that } y = \frac{\pi}{4} \text{ when } x = 1.$$

OR

(b) Find the general solution of the differential equation : *3*

$$(x + y) \frac{dx}{dy} = 1$$

9. For vectors $\vec{a} = \hat{i} + 2\hat{j} + 4\hat{k}$, $\vec{b} = -2\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{c} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, determine (i) $\vec{b} \times \vec{c}$ and hence (ii) $\vec{a} \cdot (\vec{b} \times \vec{c})$. *3*

10. (a) Find the vector and the cartesian equation of the line passing through the points A(– 1, 3, 2) and B(– 4, 2, – 2). Also, find the value of λ, if the point P(5, 5, λ) lies on the line AB. *3*

OR

(b) Find the shortest distance between the lines : 3

$$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}) \text{ and}$$

$$\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k}).$$

SECTION C

*Questions number **11** to **14** carry **4** marks each.*

11. Find : $\int \frac{x}{(x+2)(3-2x)} dx$ 4

12. (a) Using integration, find the area of the region of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ bounded by the ordinates $x = 0$, $x = 2$. 4

OR

(b) Using integration, find the area of the region of the curve $y^2 = 4x$ bounded by the ordinates $x = 1$, $x = 4$. 4

13. Find the equation of the plane passing through the points P(4, 3, 4), Q(5, 3, 1) and R(7, 6, 2). Hence, find the distance of this plane from origin. 4

14. In a class, 5% of the boys and 10% of the girls have an IQ more than 150. In this class, 60% of the students are boys. One student is selected at random from the class.

Based on the above,

(a) Find the probability that the selected student has an IQ more than 150. 2

(b) If it is given that the selected student has IQ more than 150, find the probability that the student is a girl. 2